

1. Consider scalar electrodynamics defined by the bare Lagrangian,

$$L = (D_\mu \phi)^* D^\mu \phi - m_0^2 \phi^* \phi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu},$$

where, ϕ is the unrenormalized complex scalar field, $D_\mu = \partial_\mu + ie_0 A_\mu$ and A_μ is the unrenormalized photon field. The bare quantities may be expressed in terms of the renormalized ones via the multiplicative renormalization relations:

$$\phi = \sqrt{Z_2} \phi_R, \quad A^\mu = \sqrt{Z_3} A_R^\mu, \quad m_0^2 = Z_m m_R^2, \quad e_0 = Z_e e_R.$$

Split the Lagrangian expressed in terms of the renormalized quantities in terms of the free and interaction parts, including the counterterm Lagrangian. Consider the scalar two point function at the 1-loop level. Using Dimensional regularization obtain the infinite parts of the renormalization constants Z_2 and Z_m . (We will see later in class that we need to include the term $-\frac{\lambda_0}{4}(\phi^* \phi)^2$ also in the bare Lagrangian but for purposes of this problem please ignore it.)

2. Consider QED in two dimensional space-time with massless electrons. This is called the Schwinger model. Calculate the photon self energy to one loop order. From the full photon propagator (at one loop order) read off the photon mass. Explain why your answer makes sense on dimensional grounds.